

VIDHAN JAIN

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Professional Summary

M.Tech. engineering researcher and applied AI engineer specializing in Machine Learning, Deep Learning, and Computer Vision. Experienced in designing resource-efficient neural architectures for complex perception tasks and conducting rigorous empirical model diagnosis. Proficient in Python, PyTorch, and developing applied AI solutions encompassing Generative AI, RAG pipelines, and foundational agentic workflows. Focused on bridging hypothesis-driven ML research with practical, scalable deployments to solve real-world engineering challenges.

Education

National Institute of Technology

Master of Technology in Signal And Image Processing — C.G.P.A: 9.55/10

Aug. 2024 – May 2026

Rourkela, Odisha

Jabalpur Engineering College

Bachelor of Technology in Electronics and Communication Engineering — C.G.P.A : 8.23

Aug. 2019 – May. 2023

Jabalpur, M.P.

Little World School

High School in PCM — Percentage : 88.2

April. 2018 – March. 2019

Jabalpur, M.P.

Relevant Coursework

- | | | | |
|------------------------|----------------------------|---------------------|------------------|
| • Computer Vision | • Soft Computing | Processing | • Programming in |
| • Optimization | • Linear Algebra | • Advanced DSP | C/Python |
| Techniques | • Probability & Statistics | • Data Structures & | |
| • Machine Intelligence | • Multimedia Signal | Algorithms | |

Experience

Ascendion

Distinguished Engineer

Aug. 2026 - Present

Bengaluru, Karnataka

PRISM Research Group, NIT Rourkela

PG Researcher under [Dr. Sobhan Kanti Dhara](#)

Jan 2025 – July 2026

[Website](#) — Rourkela, Odisha

- Designed and led development of **4 novel lightweight neural architectures** (TSRNet, SAPNet, CGANet, CSCA) for real-world object detection, emphasising core architectural innovation for complex vision tasks.
- Co-authored **GeoStatNet (Published, IEEE SPACE)**, **GFCR-Net (Published, IEEE GRSL)**, and an **Accepted InGaRSS** paper to advance robust model evaluation and object detection in visually degraded environments.

Space Application Centre, ISRO

Research Intern under [Dr. Deepak Putrevu](#) and [Dr. Anand S. Sahadevan](#)

June 2022 – Nov. 2022

Ahmedabad, Gujarat

- Processed large-scale airborne LiDAR datasets of complex forest canopies; developed a region-growing algorithm to delineate individual entities, solving real-world scientific problems with ambiguous, noisy data.
- Benchmarked model outputs against ground truth using rigorous statistical analysis, building the habit of measuring what matters, not just what is easy to measure.

Publications

- **Vidhan Jain**, Tanish, Sobhan Kanti Dhara, “*TSRNet: A Lightweight Thermal Saliency Refinement Network for Tiny Human Detection*”, **National Conference on Communications (NCC)**, Published, 2026.
- **Vidhan Jain**, Tanish, Sobhan Kanti Dhara, “*SAPNet: A Lightweight and Efficient Saliency-Aware Path Network for Tiny Human Detection in Thermal Aerial Imagery*”, **IEEE SPACE**, Published, 2026.
- Tanish, **Vidhan Jain**, Sobhan Kanti Dhara, “*GeoStatNet: Geometric-Statistical Feature Fusion for Robust Ship Detection in Degraded Maritime Environments*”, **IEEE SPACE**, Published, 2026.

- Tanish, **Vidhan Jain**, Sobhan Kanti Dhara, “*GFCR-Net: Gradient-Guided and Frequency-Gated Context Refinement for Robust Ship Detection in Complex Optical Remote Sensing Images*”, **IEEE Geoscience and Remote Sensing Letters (GRSL)** (Impact Factor: 5.29), **Published**, 2026.
- **Vidhan Jain**, Pratyush Sahu, Sobhan Kanti Dhara, “*Cross-Scale Correlation Guided Attention Network for Robust Tiny Human Detection in Aerial Thermal Imagery*”, **ICVGIP**, Under Review, 2026.
- **Vidhan Jain**, Pratyush Sahu, Tanish, Sobhan Kanti Dhara, “*Tiny Human Detection from Aerial Infrared Sensor Imagery via Lightweight Center-Surround Context-Aware Framework*”, **IEEE Sensor Letters**, Under Review, 2026.
- Pratyush Sahu, **Vidhan Jain**, Sobhan Kanti Dhara, “*Frequency-Aware Feature Refinement and Auxiliary Mask Supervision for Tiny Human Detection in Aerial Thermal Images*”, **IEEE India Geoscience and Remote Sensing Symposium (InGaRSS)**, **Accepted**, 2026.

Projects

Literature-RAG: Research Discovery Engine | *FAISS, SentenceTransformers, Groq LLM, Python* **GitHub**

- Built a full RAG pipeline from scratch, enforcing structural document boundaries (no naive token splitting) to preserve context integrity; embeds via **all-MiniLM-L6-v2** and indexes with **FAISS IndexFlatIP** using L2-normalised cosine similarity.
- Routes retrieved context to Groq’s Llama-3.1-8b-instant API with a pruned multi-turn memory buffer, demonstrating LLM API integration, context-window management, and retrieval-augmented inference.

Prompted Drywall Defect Segmentation | *CNN, Focal+Dice Loss* **GitHub**

- Fine-tuned an architectural segmentation network (150M params) for prompted segmentation of drywall defects, taping joints and hairline cracks, across 6,000 annotated images from two datasets.
- Addressed severe class imbalance (cracks less than 1% of pixels) using a combined Focal + Dice loss, directly optimising for spatial overlap (mIoU: 0.52, Dice: 0.67 on unified test set).
- Diagnosed and resolved a critical data pipeline issue, missing polygon annotations in the detection dataset, by engineering a bounding-box mask fallback to prevent training on blank ground truth.

Eye Disease Classification & Inference Pipeline | *PyTorch, ONNX, TensorRT, Docker* **GitHub**

- Trained ResNet50 on 5,335 medical images (84.2% accuracy) and engineered a full low-latency inference pipeline: **PyTorch** → **ONNX** → **TensorRT** — the complete path from model to production system.
- Containerised the inference environment with **Docker** for reproducible, cloud-scalable deployment.

Technical Skills

Languages: Python, SQL, MATLAB

AI/ML: Machine Learning, Deep Learning, Computer Vision, CNNs, Vision Transformers, Object Detection, Image Segmentation, Embeddings, Model Evaluation

Frameworks: PyTorch, YOLO (Ultralytics), MMDetection, TensorFlow, NumPy, Pandas

Applied AI: RAG Pipelines, LLM APIs, Prompt Engineering

MLOps: ONNX, TensorRT, GitHub, Edge Inference, Model Optimisation

Awards/Achievements

- **IndiaAI Fellowship Recipient** for M.Tech Research Project.
- **Winner** – IEEE SPACE M.Tech Student Initiative Program 2026.
- **Best Paper Award** – IEEE SPACE in SDP-1
- **Finalist** – Best M.Tech Project Award for innovative research contribution.